

Department of Engineering Cybernetics

TTK4215 Adaptive Control

Solution 12

Problem 1

The following MATLAB code implements the peak-seeking algorithm:

```
function [thetaout, etaout, xiout]=ext_seek(y, thetastart, dt)

persistent t theta eta xi omega omegal omegah k a

if isempty(t)
    % Initialize time
    t=0;

    % Initialize states of extremum seeking controller
    theta=thetastart;
    eta=y;
    xi=0;

    % Define perturbation frequency and amplitude
    omega=0.5;
    a=0.02;

    % Define cutoff frequencies for filters:
    omegal=0.05;
    omegah=0.05;

    % Define gain for update of theta.
```

```

        k=0.5;
    end

    % Extremum seeking:
    theta_dot=k*xi;
    xi_dot=-omegal*xi+omegal*(y-eta)*a*sin(omega*t);
    eta_dot=-omegah*eta+omegah*y;

    % Euler integration
    theta=theta+dt*theta_dot;
    xi=xi+dt*xi_dot;
    eta=eta+dt*eta_dot;

    % Saturation, if reasonable
    %theta=max(theta,0);
    %theta=min(theta,1);

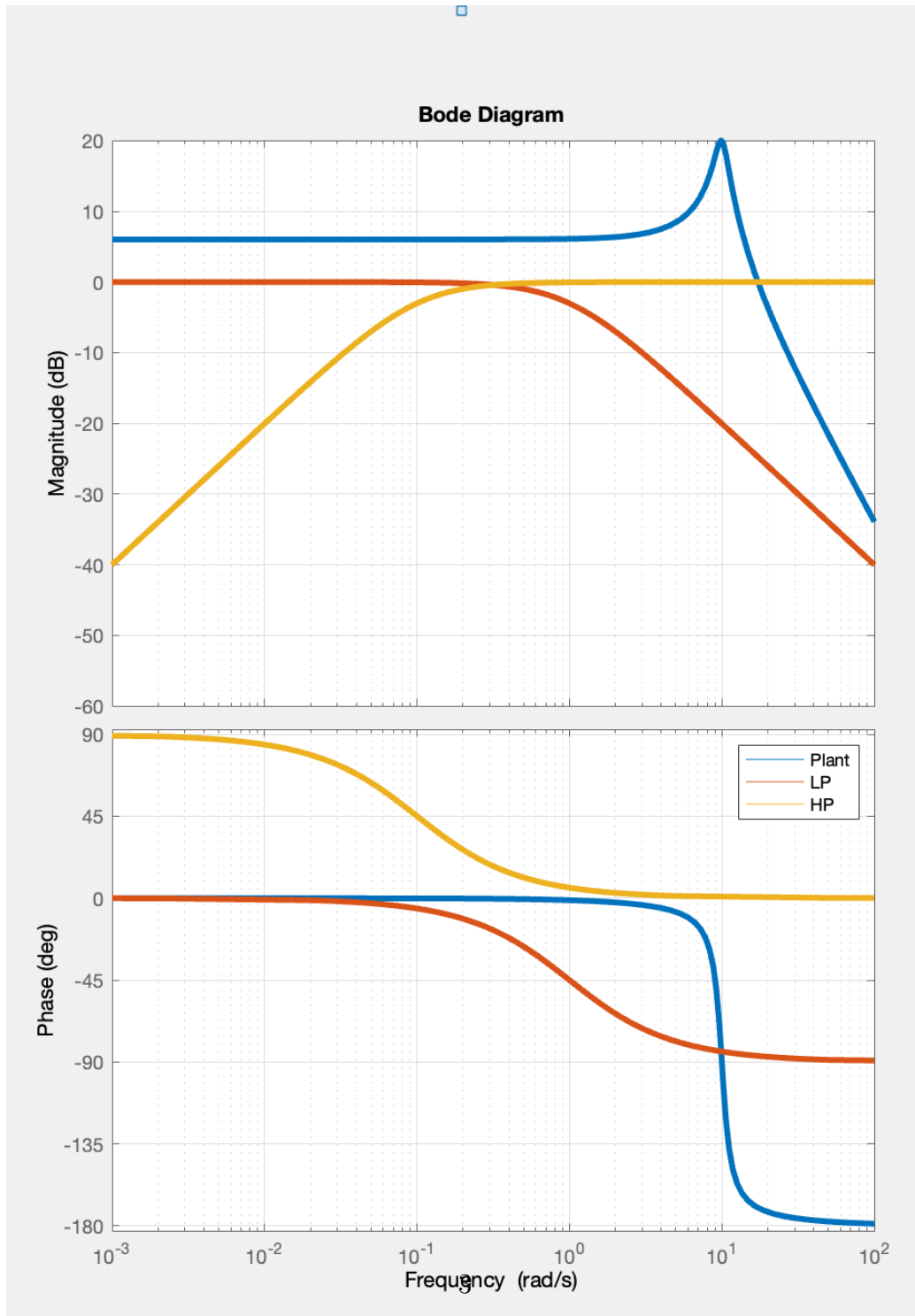
    % Return variables
    thetaout=theta+a*sin(omega*t);
    etaout=eta;
    xiout=xi;

    % Update time variable for next call to ext_seek:
    t=t+dt;

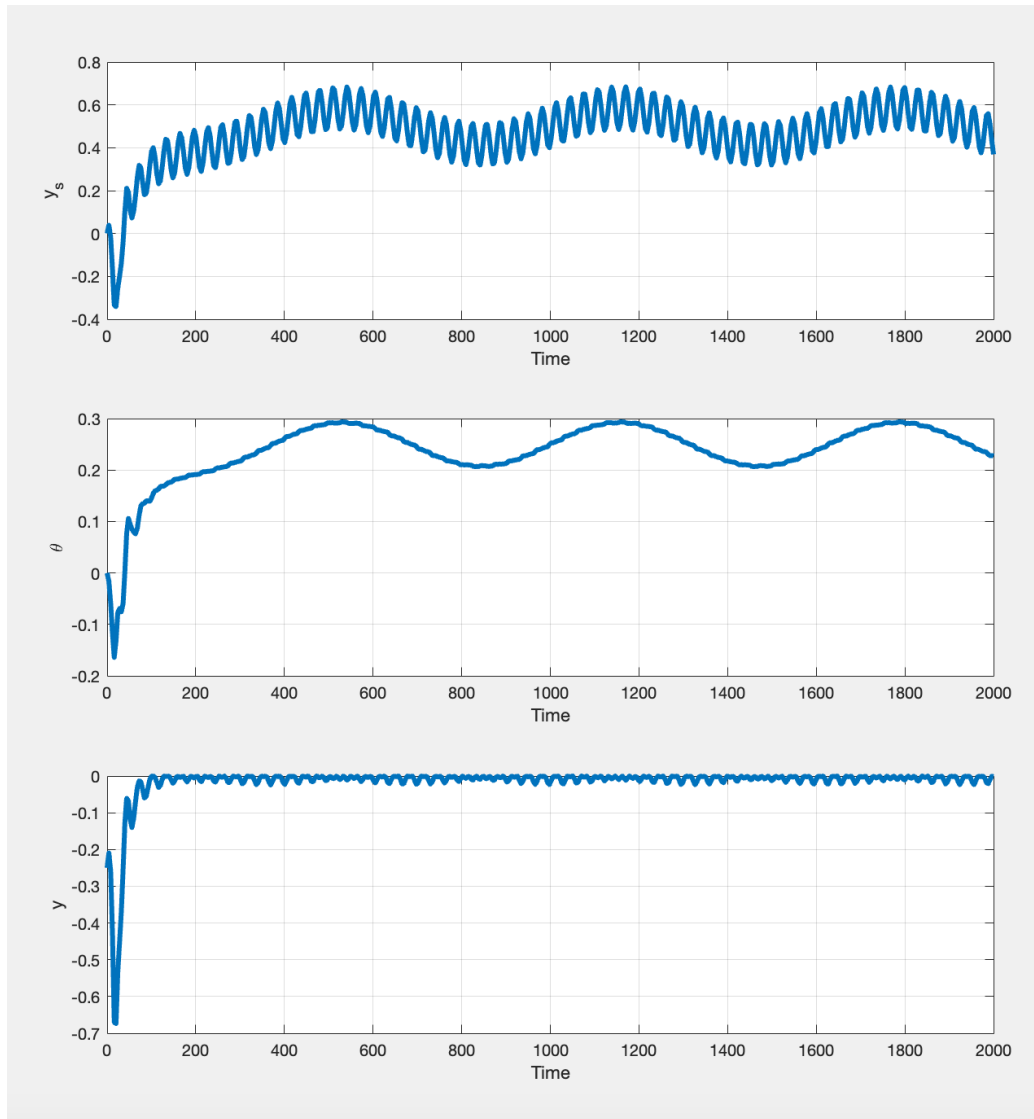
end

```

Problem 2



The low-pass filter must have cut-off frequency low enough to not excite plant dynamics. So, select it a decade less than ω_s , ie $\omega_l = 1$. The high-pass filter cut-off frequency should be selected significantly less than the low-pass filter, so select it about a decade less, ie $\omega_h = 0.1$, and the perturbation somewhere in between, so $\omega_p = 0.2$. The ESC then operates in the band-pass between the high-pass and low-pass filters. This band should be higher in frequency than the time-dependent disturbance, so that the performance index is steady from the point of view of the ESC. $\omega_d = 0.01$, so this should be ok.



Notice that the performance, y , quickly approaches its optimal value of 0 and stays there despite the periodic disturbance which requires θ to be persistently searched for.